

Enterprise Software Vendor Onboarding

AI-Assisted Intake Pipeline — Illustrated with OptiChain as the Onboarding Candidate

STATUS	OWNER	LAST UPDATED
v1.2	Procurement (Business Ops)	April 10, 2026

Pipeline: **PRD** → Design Doc → Context Contract → Agent Spec

This Product Requirements Document defines the business problem, stakeholder needs, scope, and desired outcomes for a governed AI-assisted **enterprise software vendor onboarding** workflow. Owned and version controlled by Procurement.

Target Audience: All stakeholders — procurement, legal, IT security, compliance, executive sponsors. No assumed technical knowledge

DOCUMENT LINEAGE NOTE

The four-document pipeline (PRD → Design Doc → Context Contract → Agent Spec) separates artifacts by evolution pace and audience. The PRD is stable once requirements are locked. The Design Doc changes when architecture changes. The Context Contract changes when source authority or retrieval policy changes. The Agent Spec changes most frequently, as agents are tuned. This separation is a deliberate architectural choice, not an organizational preference.

1. Executive Summary

This document defines the requirements for an AI-assisted software vendor onboarding pipeline. Lichen Manufacturing, Inc.'s evaluation of OptiChain — a demand forecasting and inventory optimization platform — has created the immediate context: the current onboarding process has no governing workflow, approvals are collected ad hoc across IT Security, Legal, Compliance, and Procurement, and the resulting cycle time of 3–5 weeks cannot meet the contract deadline. Escalations are common, and approvals are missed because requirements are fragmented across policy documents, approval matrices, system notes, and discussion threads. Teams repeatedly ask the same questions, miss required reviews, and escalate late. The enterprise wants to reduce onboarding cycle time while preserving compliance, auditability, and role-based access controls.

The pipeline defined here will replace that ad hoc process, reduce cycle time to under two weeks, and produce a complete, auditable approval record before any software vendor begins implementation.

2. Business Context & Motivation

Two conditions create immediate urgency. First, OptiChain's contract window requires a go/no-go decision within 30 days — the current ad hoc process cannot reliably meet this deadline. Second, a recent internal audit identified vendor onboarding as a compliance gap: two vendors in the prior fiscal year were granted system access before security review was complete.

WHY AI-ASSISTED

Each domain review (security, legal, procurement) currently operates in isolation, querying the same underlying policy documents independently and producing redundant, inconsistent summaries. An AI pipeline can assemble the relevant context once, route it to the right reviewer by domain, and produce a structured output that all stakeholders can act on — eliminating the coordination overhead without removing human decision authority.

3. Problem Statement

Software vendor onboarding is slow, inconsistent, and difficult to navigate because the required rules, approvals, and exceptions are scattered across disconnected sources and are never translated into a single governed workflow. Four compounding failures drive the dysfunction:

Unclear approval path	Teams cannot determine the correct approval sequence without manually cross-referencing multiple disconnected policy documents, matrices, and prior decisions.
Late discovery of required reviews	Required review steps — security questionnaires, DPA determinations, architecture reviews — are identified after work has already begun, causing rework and delay.
Exceptions and changes are buried	Recent policy changes, temporary exceptions, and informal decisions live in Slack threads and meeting notes rather than governed sources, making them effectively invisible to reviewers.
No clear next-step guidance	Business owners requesting a vendor have no way to understand why onboarding is stalled, what information is missing, or what needs to happen before the process can move forward.

4. Stakeholder Map

Stakeholder	Role	Owns	Not Responsible For
IT Security	Technical risk gatekeeping	Vendor risk classification; fast-track eligibility determination	DPA terms or legal review; budget approval
Legal	Contractual & regulatory compliance	DPA requirement determination; contract clause review	Security risk scoring; procurement path routing
Procurement	Process ownership & vendor relationship	Approval path determination; vendor questionnaire intake	Security or legal review outcomes
Compliance / Risk	Regulatory oversight	SOC2 and audit certification review; regulatory flag escalation	Contract negotiation
Operations (Business Owner)	Requesting party	Business case & justification; OptiChain use case definition	Any review domain decision

5. Scope

IN SCOPE	OUT OF SCOPE
AI-assisted security, legal, and procurement review for the OptiChain onboarding request	Vendor contract negotiation
Structured onboarding assessment: risk classification, approval checklist, DPA determination, fast-track eligibility	Post-approval implementation oversight
Retrieval from authorized internal policy sources, the vendor questionnaire, and the procurement approval matrix	Modification of the underlying IT Security Policy or DPA templates
Audit log of all AI retrieval operations, citations, and determinations	

6. Business Requirements

Requirements are ordered by dependency. A downstream requirement cannot begin until its upstream dependency is satisfied.

R-01 Vendor questionnaire intake

The OptiChain vendor questionnaire must be submitted through the formal intake process before any downstream review activity begins. No review domain may proceed without a complete submission on record.

ACCEPTANCE CRITERIA
WHEN the questionnaire has not been submitted → THE SYSTEM SHALL halt and return BLOCKED status
WHEN multiple versions exist → THE SYSTEM SHALL escalate to Procurement before proceeding

R-02 Onboarding path classification *Depends on: R-01*

The pipeline must determine the initial onboarding path for OptiChain by classifying whether the vendor’s proposed use case involves sensitive or regulated data, ERP-connected workflows, or other characteristics that increase review requirements. This classification establishes the review posture for downstream legal, security, and procurement activities and determines whether the request may proceed on a streamlined path or requires standard review. This determination gates the legal and compliance review (R-03). The fast-track eligibility and approval path determination (R-04) follows upon completion of R-03.

ACCEPTANCE CRITERIA
WHEN the OptiChain questionnaire indicates access to ERP data, production workflows, or regulated information → THE SYSTEM SHALL classify the onboarding request as requiring standard review
WHEN the OptiChain questionnaire indicates only limited use of non-regulated exported data and no production system integration → THE SYSTEM SHALL classify the onboarding request as potentially eligible for streamlined review, pending downstream domain checks

WHEN the vendor's data access, integration pattern, or intended use is ambiguous → THE SYSTEM SHALL flag the request for IT Security review before any downstream path determination is finalized

WHEN the onboarding path classification is generated → THE SYSTEM SHALL cite the authoritative policy or intake evidence used to support that classification

R-03 Legal and Compliance trigger determination *Depends on: R-02*

The pipeline must determine which legal and compliance obligations are triggered by the onboarding request based on the onboarding path classification (R-02), the vendor's data handling profile, and the legal trigger matrix. This includes determining whether a Data Processing Agreement is required and whether any contractual or compliance reviews must be completed before implementation approval may proceed.

ACCEPTANCE CRITERIA

WHEN the onboarding path classification indicates regulated or sensitive data handling → THE SYSTEM SHALL route the request to Legal review for contractual and compliance trigger determination

WHEN the legal trigger matrix indicates that a DPA is required → THE SYSTEM SHALL include the DPA as a blocker in the approval package

WHEN the legal trigger matrix indicates that a DPA is not required → THE SYSTEM SHALL cite the specific rule or trigger condition supporting that determination

WHEN legal or compliance obligations cannot be determined from the available intake information → THE SYSTEM SHALL flag missing information and return an unresolved status for human review

R-04 Fast-track eligibility determination *Depends on: R-02, R-03*

The pipeline must determine whether OptiChain qualifies for expedited approval (fast-track) or must follow the standard review path, based on IT Security's risk classification and the procurement approval matrix.

ACCEPTANCE CRITERIA

WHEN vendor handles regulated data → THE SYSTEM SHALL default to standard path, not fast-track

WHEN fast-track eligibility is borderline → THE SYSTEM SHALL default to standard path and document the rationale

WHEN fast-track is approved → THE SYSTEM SHALL cite the specific approval matrix rule

R-05 Complete approval checklist generation *Depends on: R-01, R-02, R-03, R-04*

The pipeline must produce a complete, structured approval checklist before any implementation activity begins. The checklist must identify every required sign-off, the responsible stakeholder, and estimated completion timeline.

ACCEPTANCE CRITERIA

WHEN all upstream requirements are resolved → THE SYSTEM SHALL produce a checklist with no omitted approvals

WHEN any required information is missing → THE SYSTEM SHALL list it explicitly as a blocker

WHEN output is generated → THE SYSTEM SHALL make a determination and cite an authoritative source

R-06 Stakeholder guidance and handoff support *Depends on: R-01, R-02, R-03, R-04, R-05*

The pipeline must provide each stakeholder group with actionable next-step guidance after the onboarding assessment is completed. The output must not only identify required approvals and blockers, but also organize the work so that IT Security, Legal, Procurement, and the requesting business owner can clearly understand what they are responsible for, why their review is required, and what information or action is needed to move the onboarding request forward.

ACCEPTANCE CRITERIA

WHEN the approval package is generated → THE SYSTEM SHALL identify the responsible stakeholder for each required review step, blocker, and approval action

WHEN a review step is assigned to IT Security, Legal, or Procurement → THE SYSTEM SHALL provide a concise explanation of why that review is required, supported by cited source material

WHEN required information is missing → THE SYSTEM SHALL explicitly identify the missing information, the stakeholder responsible for providing it, and the downstream step blocked by its absence

WHEN the onboarding assessment is complete → THE SYSTEM SHALL produce stakeholder-ready next-step guidance that can be used to route work without requiring the business owner to manually interpret policy documents or approval matrices

7. Constraints & Non-Negotiables

- No vendor may access internal ERP or production systems before onboarding review is complete and documented.
- Legal must sign off before any DPA is waived — the AI pipeline may recommend a waiver but may not execute one.
- Fast-track exception does not apply if the vendor touches HR data, financial records, or any PII-classified field in the ERP.
- The AI pipeline may not autonomously waive any required approval step — it may only flag, recommend, and escalate.
- Attorney-client privileged communications are excluded from all retrieval operations, regardless of relevance.

8. Risks

- The pipeline may over-rely on informal Slack threads or meeting notes unless source authority hierarchy is strictly enforced in the Context Contract.
- Stakeholders may interpret AI-generated assessments as final decisions rather than governed recommendations, reducing appropriate human oversight.
- Missing or incomplete vendor facts — particularly around ERP integration and data classification — may make accurate approval-path routing impossible without manual intervention.
- Conflicting formal policy documents will trigger mandatory escalation, reducing the automation benefit for edge cases and requiring clear human escalation paths.
- Access-control gaps in the retrieval layer could surface privileged legal communications or restricted security assessments to unauthorized reviewers.

9. Success Criteria

Metric	Target
≤ 10 business days	Total elapsed time from questionnaire submission to final approval checklist
Zero omissions	No required approvals missing from the AI-generated checklist, verified by post-process audit
Accepted without major revision	AI-generated assessment accepted by Legal and Security without requiring full re-draft
Zero compliance gaps	Zero compliance gaps attributable to this onboarding process, per internal audit

10. Open Questions

Q1: Does OptiChain's forecasting module pull from the ERP directly, or only from exported data sets?

Determines whether the ERP integration triggers the regulated data classification in R-02. If export-only, the data classification may be lower. Owner: Operations to clarify with OptiChain before Design Doc is finalized.

Q2: Is there an existing vendor relationship with OptiChain (prior pilot, NDA, or MSA) that modifies the standard approval path?

An existing relationship may change which approval steps are required under the procurement approval matrix. Owner: Procurement to confirm before R-04 can be resolved.

Q3: What is the current version status of the IT Security Policy? Is v4.2 the authoritative document?

The Context Contract will specify IT Security Policy v4.2 as an authoritative source. If a newer version exists or is in draft, the retrieval layer must be updated before the pipeline runs. Owner: IT Security to confirm.

Handoff Map — What This Document Does Not Answer

This PRD defines the business problem and what the process must accomplish. It does not specify how the system is built or how agents behave. Those questions are answered in downstream documents.

Design Doc System architecture, retrieval strategy, agent orchestration layer, data source integration, security & access control	Context Contract Which data sources are authoritative, freshness rules, context budget & prioritization, conflict resolution	Agent Spec Agent roster & roles, behavioral rules, output format, exception handling, evaluation criteria
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Version Log

Version	Date	Author	Change
v0.1 (Draft)	2026-03-31	Procurement / Business Ops	Initial draft. Scope, stakeholders, requirements, and success criteria established. Open questions flagged for resolution before Design Doc begins.
v1.1	2026-03-31	Procurement / Business Ops	Added Problem Statement (new §3) and Risks (new §8) as distinct sections. All subsequent sections renumbered.
v1.2	2026-04-10	Procurement / Business Ops	Removed parallel execution of R-03 and R-04. R-04 dependency updated from R-02 to R-02 and R-03, making the pipeline fully sequential. R-02 gating language updated to reflect that R-02 gates R-03, and R-03 completion gates R-04. Cover version and date updated.